

LNCT UNIVERSITY, BHOPAL
CSE-604
B.TECH (CSE) VI SEMESTER
Design Thinking and Innovation(CS604)

Maximum Marks: 70

Time Allowed: 3 Hours

Note:- Attempt all questions. Internal choice is given.

SECTION – A

I. Short Answer Type Questions (Attempt Any Five) [5×4=20]

- i. What makes Design Thinking different from traditional problem-solving approaches?
- ii. List any four principles of Design Thinking with one-line explanations.
- iii. Define observation and explain its role in empathizing with users.
- iv. What is brainstorming? Mention any two ground rules to conduct it effectively.
- v. Differentiate between primary research and secondary research in design thinking.
- vi. What is a storyboard or scenario in storytelling? How does it help in prototyping?
- vii. Define entrepreneurship. Mention any two common challenges faced by new entrepreneurs.

SECTION – B

II. Long Answer Type Questions (Attempt Any Four) [4×10=40]

- i. Explain the **process of Design Thinking**. Illustrate it with an example project or case study.
- ii. Discuss the use of **empathizing techniques** like structured open-ended interviews and behavioral models. Provide a case-based example.
- iii. Explain the role of **Empathy Maps, Mind Maps, Affinity Diagrams, and Journey Maps** in design thinking.
- iv. Discuss **prototyping techniques (soft, medium, final)**. How do these support innovation and usability studies?
- v. Explain the **engineering aspects** in design with relevance to material, mechanical, and safety considerations.
- vi. Describe the process of launching a **startup using a design-thinking and entrepreneurial approach**. What factors help make it successful?